



Mo	Tu	We	Th	Fr	Sa	Su
----	----	----	----	----	----	----

Memo No. _____

Date

/ /

Week 4 Day 1

Data Base Testing

It is a software testing process used to verify the accuracy, integrity, consistency, and reliability of data stored in a database.

Types of DB Testing

1) Structural Testing

It checks the structure and internal parts of a database.

a) Schema Testing - It checks the structure of the database is correct or not.

b) Table & Column Testing - It checks the table name, column names, data types and field sizes.

c) Database Server Validation - Checks server ~~st~~ setting, permissions, and ability to handle users.

2) Functional Testing

It checks whether the database works according to the requirements

a) BlackBox Testing

Test database functions without

looking at the internal structure.

b) White Box Testing

Tests internal database logic, triggers, views and procedures.

c) Non-functional testing

It checks the performance, security, usability, and compatibility with other systems.

Types of Data

1) Structured Data

The data that is organized in a fixed format. for eg. Student records, employee details.

2) Unstructured Data

The data that does not have a fixed format or structure. for eg. Img, vdo, audio, email, social media post.

3) Semi-structured Data

The data that does not have a fixed format but have some organization for eg: JSON, XML, and HTML files.

Database components

- 1> Database Schema
- 2> Transaction
- 3> Triggers
- 4> Stored Procedures
- 5> Field constraints

Objectives of Database Testing

1> Data Mapping

It ensure that the frontend data fields are correctly connected to database tables and CRUD operations works properly

2> ACID Properties . .

It ensure that the database transaction are reliable and accurate.

3> Data integrity

It ensure that the data is correct, complete, and consistent everywhere.

4> Business rule accuracy .

It checks the rules triggers, and stored procedures work correctly.

ACID Properties

They ensure the transactions of database are reliable and accurate.

i) Atomicity

It means a transaction is all or nothing either all its operations succeed or none are applied. It follow commit or Rollback.

for eg

Before $X = 500$ $Y = 200$	
Transaction T	
T1	T2
Read (X)	Read (Y)
$X := X - 100$	$Y := Y + 100$
Write (X)	Write (Y)
After $X = 400$	$Y = 300$

ii) Consistency

A transaction must keep the database correct and valid before and after the transaction. For eg- If \$100 is transferred from A to B the total supply money should remain the same.



Mo	Tu	We	Th	Fr	Sa	Su
----	----	----	----	----	----	----

Memo No. _____

Date / /

iii) Isolation

Multiple transaction can run at the same time without interfering with each other.

iv)

Durability

Once a transaction is committed / save the change remainly saved even if the system crashed. For eg. - If the money transfer was successful, the updated balaces remains even if the computer suddenly shut down.

DDL (Data definition Language)

DDL is a set of SQL commands used to define, create, and modify the structure of database such as tables, schemas, views, and indexes. CREATE, ALTER, DROP, TRUNCAT

DML (Data Manipulation Language)

It is used to add, retrieve, update, select, Merge, delete.

Data type

- Int
- Decimal
- Char
- Varchar
- Data
- Time
- Boolean
- Blob

Constraints in DBMS

i) Primary Key

Unique identifiers each record in a table. It cannot be null for eg. Student-ID.

ii) Foreign Key

Connects one table to another and ensure valid relationships. Eg. Course-ID in a student table.

iii) Unique

It ensure that all values in the column are different.

iv) Not Null

It ensure a column cannot be left empty.